What's Left to Be Done - Quick Reference

Last Updated: January 2025
Overall Completion: ~65%

🎯 Quick Status

☑ Production-Ready (5 Engines)

- **Data Integration** (95%) - Weather data ingestion
- **Temporal Logic** (90%) - Task scheduling
- **Geo Context** (85%) - Spatial queries (Poland/Zimbabwe)
- **Image Analysis** (90%) - Crop disease detection
- **AgroLLM** (95%) - Safety validation, recommendations

⚠ Needs Work (2 Engines)

- **Conversational NLP** (70%) - Uses stub backend, needs real LLM
- **Crop Intelligence** (75%) - 4-6 weeks to complete

❌ Not Implemented (12 Engines)

- Inventory, Diagnosis Recommendation, and 10 other skeleton engines

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🔴 CRITICAL - Fix Before Production

Security Issues (1 week)

1. **SQL Injection** - `data_integration/storage/db.py:72` (2-4 hours)
2. **JWT Bypass** - `shared/security/deps.py:44-49` (1-2 hours)
3. **Error Leakage** - Multiple endpoints (1 day)
4. **Missing** `.env.example` - Create templates (1 day)
5. **Rate Limiting** - Apply to API gateway (2-3 days)

Infrastructure (2-3 weeks)

- ☐ CI/CD pipeline (GitHub Actions)
- ☐ Monitoring alerts (Prometheus)
- ☐ Log aggregation (ELK/Loki)
- ☐ Database backups
- ☐ Redis cluster (remove SPOF)

🟠 HIGH PRIORITY - Before Launch

Core Features (8-10 weeks)

1. **Complete Crop Intelligence Engine** (4-6 weeks)
    - Integrate FAO-56 water management
    - Implement GDD calculations
    - Verify/obtain ML model artifacts
    - Budget filtering
    - Disease risk expansion
2. **Real LLM Integration** (2-3 weeks)
    - Replace stub backend in Conversational NLP
    - Integrate actual LLM model
    - Test end-to-end
3. **Billing System** (2-3 weeks)
    - Usage tracking/metering
    - Subscription management
    - Payment integration

Testing (2 weeks)

- ☐ Increase test coverage to >80%
- ☐ Load testing (10K concurrent users)
- ☐ Integration tests
- ☐ Security testing

🟢 MEDIUM PRIORITY - Post-Launch

Regional Expansion

- ☐ Add 3-5 more regions
- ☐ Region-specific regulations
- ☐ Multi-language support

Skeleton Engines

- ☐ Prioritize by business need
- ☐ Inventory Engine (high value)
- ☐ Diagnosis Recommendation (high value)
- ☐ Others as needed

Technical Debt

- ☐ Resolve 136 TODO/FIXME comments
- ☐ Clean up `*_old.py` files
- ☐ Standardize error handling
- ☐ Add circuit breakers

📅 Timeline to Production

Phase	Duration	Tasks
Phase 1: Security	1 week	Fix critical vulnerabilities
Phase 2: Core Features	8-10 weeks	Complete Crop Intelligence, LLM integration
Phase 3: Infrastructure	2-3 weeks	CI/CD, monitoring, backups
Phase 4: Testing	2 weeks	Coverage, load testing
Total	**12-16 weeks**	With 2-3 engineers

🎯 Immediate Next Steps

1. **This Week**: Fix all critical security issues
2. **Next 2 Weeks**: Set up CI/CD and basic monitoring
3. **Next 8 Weeks**: Complete Crop Intelligence Engine
4. **Next 2 Weeks**: Add billing system
5. **Next 4 Weeks**: Run pilot with 10-20 farmers
6. **Then**: Seek investment with traction data

📊 Completion Checklist

Security ☑/❌

- ☐ SQL injection fixed
- ☐ JWT bypass removed
- ☐ Error leakage fixed
- ☐ `.env.example` files created
- ☐ Rate limiting applied
- ☐ Dependency vulnerabilities resolved

Features ☑/❌

- ☐ Crop Intelligence Engine complete
- ☐ Conversational NLP with real LLM
- ☐ Billing system implemented
- ☐ Usage tracking/metering
- ☐ At least 3 regions supported

Infrastructure ☑/❌

- ☐ CI/CD pipeline
- ☐ Monitoring & alerts
- ☐ Log aggregation
- ☐ Database backups
- ☐ Redis cluster

- ☐ Load balancer

Testing ☑/❌

- ☐ Test coverage >80%
- ☐ Load testing complete
- ☐ Security testing complete
- ☐ Integration tests passing

Documentation ☑/❌

- ☐ Architecture diagrams
- ☐ API documentation
- ☐ Operational runbooks
- ☐ Disaster recovery plan

See `INVESTOR_CODE_AUDIT.md` for detailed analysis